

Patient Clinical Note for Pathology Requisition:

Summary of Request:

I have an 11-year stable, isolated NRAS+ thyroid nodule (TR3) with no other mutations and one nodule in lower right pole (TR4) with abundant background colloid and normal lymph nodes. The previous indeterminate/microfollicular/Hürthle-predominant result was sampled during an acute, localized phototoxic surface burn and a regional dental infection (systemic, sepsis symptoms). This repeat biopsy is at 5 weeks post-resolution to evaluate for the clearance of reactive/degenerative atypia. In addition, the molecular test is severely unreliable: it has a 53% false positive rate according to Columbia University [1]. This hypothesis is fascinating, and I truly want RFA to save my thyroid (especially given that autistic people have sensitivities to ingredients and medications, and I would like to avoid being on hormone replacement at all costs).

Detailed Analysis

This document presents a systems-biology analysis of the localized cellular environment surrounding my thyroid nodules. It explores the relationship between photobiomodulation-induced mitochondrial stress, systemic inflammatory markers, and the resulting transcriptomic 'noise' that may have led to the indeterminate molecular result in my recent biopsy. This is provided to assist in the cytopathological re-evaluation of my case. My endocrinologist and interventional radiologist at UCLA also agrees with this approach (that a repeat biopsy is needed to clarify things due to the massive systemic stress that likely caused reactive atypia, and that NRAS mutations can be completely indolent/benign, like my first biopsy showed; this will clear me for RFA to resolve the significant compressive symptoms I am experiencing with these nodules).

History: Two thyroid nodules with 11-year documented photographic stability (isoechoic; no lymphadenopathy). One has peripheral rim calcification (TR4) in recent ultrasound, but this more concerning one is avascular, supporting the hypothesis that this is a benign 'mummified' nodule.

Systemic Context:



Figure 1: Photo from June 2015 prior to the start of the Charlevoix marathon back when I lived in Michigan, when tensing neck due to nervousness, unidentified larger nodule (mid right pole) became prominent, looks to be around the same as it is today)

June 2015: Noticed unidentified nodules in photo shown above, showing they were there 11 years ago (in 2025, looked back through photos for evidence of them).

February 2025: Felt nodules in neck, ultrasound showed 2.5-2.8 cm isoechoic TR3 nodule in right mid pole, 1.4-1.5 cm nodule in lower right pole TR5

February 2025: Biopsy done at Olive View with cell block: both benign bland-appearing follicular cells arranged in sheets and clusters. More concerning hypoechoic at the time TR5 nodule's sample was hypercellular but bland, indicating the bland architecture persisted throughout the hypoechoic sonographic architecture.

Microscopic examination performed.

The smears are moderately cellular and show clusters and sheets of bland-appearing follicular cells. Colloid noted in the bloody background. Sections from the cell block show scant follicular cells and concur.

Comment

Performing Laboratory Information:

Smears are hypercellular and show sheets and clusters of bland-appearing follicular cells. Pigment-laden macrophages and colloid noted in the bloody background. Sections from the cell block shows mostly blood. Colloid, minute fragment of stromal tissue and scant follicular cells noted.

Figure 2: Excerpt from first benign biopsy on both nodules (Image 1: TR3 mid-pole moderately cellular nodule, Image 2: TR5 (now TR4) lower right pole hypercellular nodule) Source: Pathology Report at UCLA Olive View February 2025

December 2025-January 2026: Significant bacterial pneumonia/pleural effusion, massive systemic inflammation for months.

February 2026: Began using red light therapy as directed (especially near thyroid), as I read articles showing it improved thyroid health and reduced inflammation.

March 2026: Noticed giant weeping serous burn/rash over thyroid (and on back of neck), not sure what it was from. Thyroid felt inflamed and throat was very sore. This burn appeared directly on the side that received the most stimulation from the red light.

March 2026: Major dental surgeries (implant on lower right molar, wisdom tooth extracted lower left with massive infection, and complex block bone graft at tooth #10 (front of mouth top tooth). Resolved after a few weeks with antibiotics, lots of transient postoperative swelling and inflammation.

Ultrasound UCLA health late March 2026: ultrasound showed 2.7 cm isoechoic TR3 nodule in right mid pole, 1.6 cm nodule in lower right pole downgraded to TR4 (TTW to WTT and increased in echogenicity, both nodules stable in size according to endocrinologist and interventional radiologist). Burn with pustules and itchiness present at ultrasound

April 2026: Rash got worse, diagnosed as complex burn with secondary staph, antibiotic cream started. UCLA Health urgent care diagnosed me with complex burn with secondary staph. Noticed significant correlation with red light therapy. Now know that this rash is likely a phototoxic dermatitis event/thermal burn from the red light device or polymorphous light eruption-type reaction with possible secondary staph. Massive rash over thyroid and thyroiditis symptoms (swollen feeling near nodules, constant sore throat and inflammation, nodules felt enlarged when touching them, skin burns and skin feeling leathery and peeling on neck) due to this red light.



Figure 3: Localized phototoxic burn and potential secondary infection above thyroid on 2026-04-17 (11 days before second biopsy), illustrating acute inflammatory state.

April 17 2026: Traumatic wisdom teeth extraction, badly impacted, massive abscess and systemic infection, neurological pain, pustules all over that side of my face and back, sepsis symptoms (severe confusion, fast heartrate, lower BP than usual). Same day as photo of rash taken above. Took weeks to fully resolve with amoxicillin.

April 28 2026: Second biopsy showed oncocytic changes and Bethesda III (AUS); hypothesize this was a snapshot of reactive atypia (oncocytic metaplasia) from this “perfect biological storm,” not the baseline phenotype. Red light therapy is known to induce mitochondrial biogenesis (leading to potential pseudo-oncocytic changes), and it seemed the device was very faulty for it to have caused such a significant, months-long thermal phototoxic burn. Note that I do not have Hashimoto’s antibodies and have a homogenous echotexture and no lymphocytes present in my bland “baseline state” initial biopsy done at Olive View, further suggesting this may have been a reactive response.

A. THYROID, RIGHT, MID, NODULE (US-GUIDED FINE NEEDLE ASPIRATION):

- Atypia of undetermined significance, oncocytic type
- Background suggestive of lymphocytic thyroiditis

B. THYROID, RIGHT, INFERIOR, NODULE (US-GUIDED FINE NEEDLE ASPIRATION):

- Atypia of undetermined significance, oncocytic type

COMMENT: Specimen will be submitted for molecular testing and results will be posted separately. Please correlate cytology findings with molecular results.

THYROID BETHESDA CATEGORY

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Specimen A - III. Atypia of undetermined significance

Specimen B - III. Atypia of undetermined significance

MICROSCOPIC DESCRIPTION

A. Cellular smears consist of small groups and clusters of follicular and predominantly oncocytic cells presenting disorganized, pleomorphic round to oval round to oval nuclei with mildly irregular nuclear contours, fine or pale chromatin, rare nuclear grooves, small nucleoli and fine or dense cytoplasm. Increased micro-follicles, lymphocytes are seen and colloid are noted in a bloody background. ThinPrep prepared and concurred. Adequacy performed by cytotechnologist (SB/YH).

B. Cellular smears contain follicular cells and Hurthle cells arranged in cohesive sheets, micro follicles and small groups of small round to oval nuclei, fine or pale chromatin, delicate or dense cytoplasm in a bloody background with colloid. ThinPrep prepared and concurred. Adequacy performed by cytotechnologist (SB/YH).

Figure 4: Excerpt from second indeterminate biopsy on both nodules, showing atypia consistent with the gradient hypothesis below (Source: UCLA Health Pathology Report, April 2026)

I also want to add that I noticed a spatial-temporal correlation with the atypia "gradient" seen in my nodules.

Nodule A: (Proximity Effect): Located directly under the burn site (larger nodule, mid right pole), this nodule exhibited the most significant reactive changes: pleomorphic, disorganized nuclei, lymphocytic infiltration, and Hürthle cell dominance. This represents the high-intensity zone of the inflammatory gradient, where the systemic inflammatory load from the dental infection and the localized phototoxic burn intersected maximally.

Nodule B: (Gradient Effect): Located further away (lower right pole), this nodule showed less reactive atypia, as it was partially shielded from the peak intensity of the inflammatory insult. The presence of Hürthle cells in cohesive sheets rather than disorganized pleomorphism suggests a lower-grade reactive response.

Several benign-leading features remain despite the atypia: presence of colloid, and Nodule B being organized in "cohesive sheets." Also, ThinPrep has been known to obscure colloid, which may have contributed to the atypia appearance, as it isolates cells (that's why I'm requesting a cell block for my repeat biopsy, to see the 3D architecture of my nodule and background colloid context thoroughly).

This hypothesis is fascinating, and I truly want RFA to save my thyroid. Crucially, the prior UCLA health biopsy occurred during a period of acute systemic inflammation (dental abscess/sepsis) and chronic localized thermal stress (RLT/suspected faulty device that was emitting a lot of heat and unusually bright light).

Current (May 2026):

- The dental infection completely resolved weeks ago.
- The burn has entered a remodeling phase with scar tissue (no active pustules, no itching, no burning, significantly less redness)
- I will be seeing a dermatologist next week to further evaluate the burn (the wait time was long for appointments, I would have gotten in sooner if I could have).
- The thyroiditis symptoms I was noticing since the red light burn and infection have calmed down.

Core Hypothesis: The previous "indeterminate" cytology and molecular signature likely reflect reactive oncocytic metaplasia and transcriptomic stress responses secondary to these transient systemic and thermal stressors, rather than intrinsic neoplasia.

I sincerely believe the indeterminate result was due to the aforementioned months-long thermal/phototoxic burn from red light therapy (defective device) applied to the thyroid (hoping it would shrink nodules and stupidly inspired by the studies showing phototoxic reactions with low-level laser therapy could shrink tumors), a staph infection over the burn (with pustules at the time of biopsy), and a massive dental abscess on my lower right wisdom tooth (with early sepsis symptoms) created a perfect storm (reactive oncocytic metaplasia).

NRAS codon 61 mutations are present in many benign adenomas [2], and my RNA must have looked 'noisy' due to these systemic factors. My first biopsy showed bland-appearing follicular cells in both nodules. It's rare for both nodules to transform into a neoplasm, but reactive atypia absolutely would happen if they were both subjected to the same radiant and thermal energy causing phototoxic burns coupled with a severe, systemic infection. Also, interestingly, red light therapy stimulates unusual mitochondrial biogenesis, and oncocytic changes are characterized by Hürthle cells with abundant mitochondria. Due to this hypothesis, I am requesting a repeat third 'tie-breaker' biopsy (with this entire clinical context considered) and to pursue RFA if that one comes back benign. I would really like to avoid surgery, and RFA would be the perfect solution that aligns with my conservative tissue-sparing approach. The 11-year photographic stability further supports the hypothesis that this is an indolent growth.

Given that my inflammatory state has improved, I would like to request a repeat FNA with a dedicated cell block to allow for better histological assessment of the tissue

architecture. Dellal Kahramanca, F. D., et al. (2024) recently conducted a study on patients undergoing fine needle aspiration biopsy (FNAB) for thyroid nodules, and found the rate of adequate or atypia of undetermined significance (AUS, which I have) cytology results at rebiopsy did not vary significantly with the time of repeat biopsy, suggesting a third biopsy could be repeated at 1 month or even earlier [3].

References:

1. Columbia Surgery. (2015, January 30). *Molecular testing for thyroid nodules*. Columbia University Irving Medical Center, Department of Surgery. <https://columbiasurgery.org/news/2015/01/30/molecular-testing-thyroid-nodules>
2. Wong, L. Q., & Baloch, Z. W. (2026). Impact of Co-Mutations and Genetic Variations on Malignancy Risk in RAS-Positive Indeterminate Thyroid Nodules: an Institutional Experience. *Endocrine pathology*, 37(1), 13. <https://doi.org/10.1007/s12022-026-09910-6>
3. Dellal Kahramanca, F. D., et al. (2024). Timing of the repeat thyroid fine-needle aspiration biopsy: does early repeat biopsy change the rate of nondiagnostic or atypia of undetermined significance cytology result?. *Endocrine*, 86(3), 1065–1072. <https://doi.org/10.1007/s12020-024-03953-7>